Cartoonize the Ride! Segmentation-Gated Video Cartoonization for In-car Modified Reality

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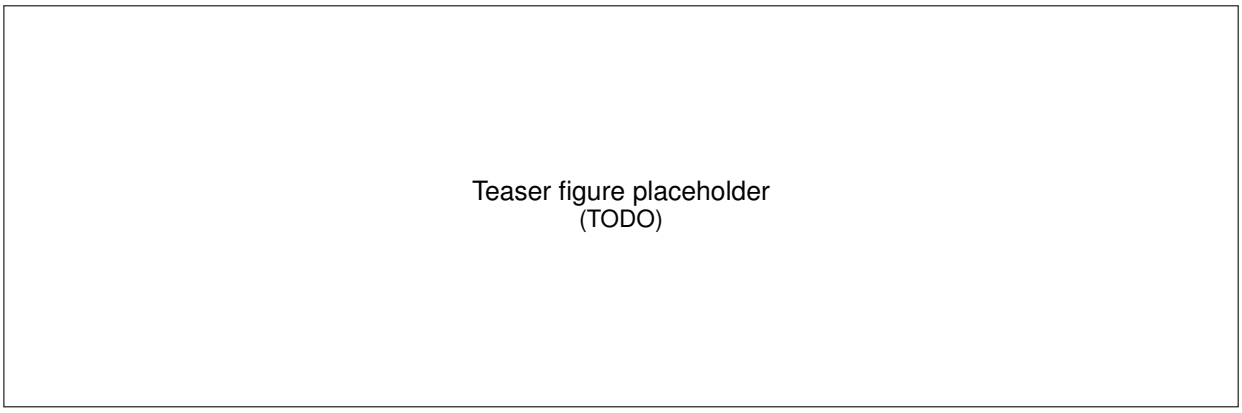


Figure 1: TODO. Teaser figure showing a passenger's view of a selectively cartoonized ride.

Abstract

As automated vehicles ease people out of the driving task, passengers can turn the ride itself into a source of entertainment, and in-car media can benefit from reframing, rather than merely displaying, the driving environment. As a step toward cartoonizing a vehicle's surround-view camera stream for its passengers, we present a segmentation-gated video cartoonization pipeline for in-car modified reality that selectively cartoonizes semantic classes in real 360° driving footage. Road users, infrastructure, or the surrounding scene can be translated into cartoon media or kept photorealistic, so that the transformation entertains while supporting situation awareness. Because the resulting media is a direct translation of the actual driving scene, it natively preserves the visual–motion congruence and road context that in-car XR media otherwise must reconstruct, and the traffic situation that explains every stop and turn remains part of the content itself. Our cascade shares the expensive per-frame computations of instance segmentation, full-frame cartoonization, and optical flow across all outputs, generating ten "reality presets" spanning a design space of selective cartoonization in a single pass. Temporal stability is maintained through flow-warped mask propagation with exponential smoothing, and boundaries between cartoonized and photorealistic regions are refined with edge-aware guided filtering. As a proof of concept, we evaluate temporal consistency, boundary quality, and throughput on real footage, deploy the resulting presets in a Unity-based in-car XR platform, and analyze the remaining gap to live, on-road use.

Index Terms: In-car entertainment, modified reality, video cartoonization, semantic segmentation, 360° video.

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1 Introduction

Commuting occupies a substantial portion of daily life [4], and the transition toward automated vehicles reframes this time. As drivers become passengers, they can engage in a broader range of non-driving-related tasks [16], and the cabin emerges as an interaction space for immersive media [6, 26], spanning information delivery [5], productivity [23], and richer sensory couplings between vehicle motion and virtual content [13, 12, 14, 31, 28].

A persistent constraint on such media is that, to avoid discomfort caused by visuo-vestibular mismatch, content must align with the actual road path and unfold at the vehicle's pace [22, 3]. Recent work relaxes this coupling to grant users freedom [28, 13, 12], yet alignment remains central to comfort [7, 9, 25], so in-car XR media are frequently built on manually constructed, route-specific pipelines that contradict the medium's mobile nature. An alternative is to derive the media from the driving scene itself. Translating the live camera view into an animated world inherently reflects the vehicle's trajectory and the surrounding traffic, offering visual–motion congruence and situation awareness without dedicated authoring [11]. Appearance-level transformation of the driving view has been identified as a mediation effect for vehicles [17] and is becoming practical through generative models for XR media [33]. Applied uniformly to the whole frame, however, it offers no control over which parts of reality are transformed, even though abstracting safety-relevant actors can impair the interpretability passengers rely on [17, 20] and emerging Automotive Mediated Reality (AMR) frameworks explicitly call for object-level mediation [17].

In this paper, we ask what it means to cartoonize a drive *selectively*. We envision a vehicle whose roof-mounted 360° camera stream is continuously translated into an animation-like world and presented to passengers in real time, with the target of translation chosen per semantic class. Only the surrounding vehicles become cartoons, or only the pedestrians, or everything except the road users, or the entire scene. Such class-selective cartoonization instantiates the *modified reality* (ModR) category of AMR effects, which modifies the appearance of existing scene elements rather than adding or removing them [17], and spans the trade-off between situation awareness and immersion that in-car XR research identi-

fies as central [20].

As a step toward this vision, we contribute *Cartoonize the Ride*, a segmentation-gated pipeline that transforms real 360° driving footage into a family of *reality presets* by gating a full-frame cartoonization model with temporally stabilized, class-specific soft masks. Because a passenger ultimately watches one reality at a time, the practical authoring question is which selective reality suits which experience. The pipeline therefore shares the expensive per-frame computations of instance segmentation, full-frame cartoonization, and optical flow across all presets, generating ten reality presets in a single pass. The resulting videos play back directly on MAXIM [30, 29], a 360° video-see-through in-car XR platform with vehicle-compensated head tracking [19]. Our contributions are threefold. First, we articulate class-selective video cartoonization as a ModR approach and organize it into a taxonomy of ten reality presets over road-relevant semantic groups. Second, we present a single-pass, computation-sharing pipeline realizing this taxonomy on real 360° footage, combining flow-warped mask smoothing with edge-aware boundary refinement. Third, we report a proof-of-concept evaluation of temporal consistency, boundary quality, and throughput, and analyze the remaining gap to real-time on-road deployment.

2 Related Work

2.1 In-car XR and Automotive Mediated Reality

Automated vehicles allow passengers to immerse themselves in alternate spaces [6], and in-car XR systems have steadily expanded the sensory and agency envelope of the ride. AttraCar repurposes built-in vehicle systems for thermal, airflow, and motion feedback [31], force mappings reinterpret vehicle dynamics as diegetic environmental feedback [14], contextual cues enable user-applied locomotion within a physically constrained ride [12, 13], and passive motion gains extend perceived motion beyond the vehicle's trajectory [28]. Common to these systems is the need for content whose motion remains plausible with respect to the vehicle, a requirement our approach satisfies natively since the media is a translation of the ride itself.

A complementary perspective is offered by Automotive Mediated Reality. MIRAGE conceptualizes AMR as a mediation layer over the driving environment and organizes its effects into augmentation, diminishment, and modification, listing appearance-level transformations of existing objects as ModR effects [17]. Our work populates this ModR region with a concrete, class-selective cartoonization instrument focused on high-quality, temporally stabilized output for entire drives. In parallel, generative approaches are beginning to re-synthesize the driving view itself. SEGAR couples a diffusion-based world model with selective correction so that augmented future frames can be generated ahead of time while safety-critical regions are realigned with reality [1]. Such generative AR shares our premise that region-selective mediation is the appropriate unit of control, and our pipeline offers a deterministic, video-translation counterpart whose outputs remain pixel-anchored to the captured footage.

2.2 Video Cartoonization and Semantic Gating

Image-to-animation translation has matured from generic image translation [32] into dedicated cartoonization models. White-box cartoonization decomposes the task into surface, structure, and texture representations [27], and the AnimeGAN family provides lightweight generators for photo animation [21]. Applied frame by frame, such models exhibit temporal flicker, which prior work links to visual demand and visually induced sickness in immersive viewing [2]. Our pipeline instead stabilizes the *gating masks* with flow-warped exponential smoothing [10] and uses a deterministic feed-forward cartoonizer at a fixed processing resolution, confining residual instability to the style domain rather than to the composition boundary.

Semantic gating requires per-class masks at video rate. We build on real-time instance segmentation from the YOLO family [18], whose COCO vocabulary covers the traffic-relevant classes in our presets, and refine mask boundaries with guided filtering [15]. Composition between two different appearance domains, cartoon and photograph, raises boundary requirements beyond conventional matting. Hard masks expose jagged, style-discontinuous seams, whereas naive feathering bleeds cartoon appearance beyond the object. Our refinement combines guided filtering with a smoothstep re-sharpening so that object interiors remain fully cartoonized while only a narrow band blends the two domains.

3 Cartoonize the Ride

3.1 Reality Presets for Selective Cartoonization

We define a *reality preset* as an assignment of appearance domains (cartoon vs. photorealistic) to semantic regions of the driving scene. Guided by the AMR design space [17] and the classes most consequential for traffic interpretation, we instantiate five mask groups, namely *person*, *vehicles* (bicycle, car, motorcycle, bus, truck, train), *person+vehicles* (road users), *all instances* (things), and *road infrastructure* (traffic lights, signs, hydrants), and derive ten presets. For each group, a *toon* variant cartoonizes the group, a *real* variant cartoonizes everything except the group, and a *full_toon* baseline cartoonizes the entire frame. Pairs share one mask, so ten presets require only five groups. The taxonomy spans complementary experiential hypotheses. *road_users_real* immerses the passenger in an animated world while keeping safety-critical actors photorealistic, aligned with concerns about semantic clarity in mediated driving scenes [17, 20], whereas *pedestrians_toon* abstracts precisely those actors into the animated domain.

3.2 Pipeline

Figure 2 summarizes the pipeline, whose organizing principle is that expensive per-frame computations are executed once and shared by all presets, while per-preset cost reduces to alpha compositing.

Shared per-frame computation. Each frame F_t is processed by three parallel branches. (i) An instance segmentation model (YOLO11n-seg [18]) produces per-instance masks and class labels. (ii) A full-frame cartoonization model (AnimeGANv3 [21]) produces the cartoonized frame C_t, with the frame resampled to a fixed short-side resolution of 1280 before inference and upsampled back, which bounds inference cost and stabilizes style granularity across frames. (iii) Dense backward optical flow [10] is estimated at quarter resolution between F_t and F_{t-1}, yielding a warp field that transports the previous frame's masks into the current frame.

Shared mask groups. For each mask group g, the union of its instance masks forms a fresh mask M_t^g. To suppress frame-to-frame mask flicker, that is, detection dropouts and boundary jitter that would otherwise appear as popping between appearance domains, each group maintains a temporally smoothed mask

$$\tilde{M}_t^g \;=\; \beta\, M_t^g \;+\; (1-\beta)\, \mathscr{W}_t\big(\tilde{M}_{t-1}^g\big)\,, \tag{1}$$

where $\mathscr{W}_t$ warps the previous smoothed mask along the estimated flow and β=0.6. On frames where segmentation is skipped (every second frame), the warped mask alone propagates the group state. The smoothed mask is refined by a guided filter [15] with the frame as guidance, followed by a smoothstep re-sharpening that restores full opacity in object interiors while retaining a narrow soft boundary transition. Refinement is computed once per *group* and reused by both presets of the pair.

Per-preset compositing. Each preset p selects its alpha map A_t^p as the refined group mask for *toon* variants, its complement for *real*

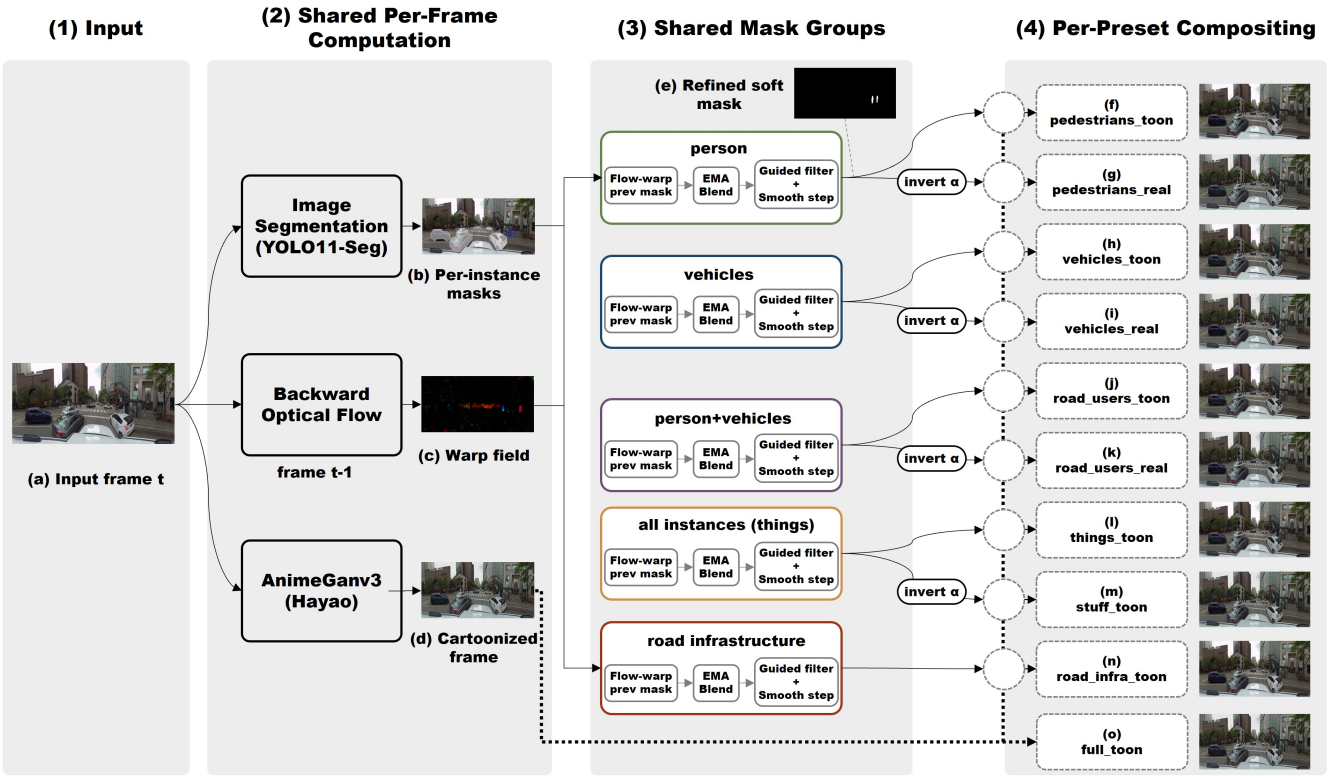


Figure 2: Pipeline overview. Each frame (a) feeds three shared computations executed once per frame, producing per-instance masks (b), a warp field (c), and a cartoonized frame (d). Five shared mask groups stabilize and refine class masks through flow-warped EMA blending and edge-aware filtering (e), and per-preset alpha compositing, inverted for the *real* variants, yields ten reality presets (f–o).

variants, or the constant 1 for *full_toon*, and composites

$$I_t^p = A_t^p \odot C_t + \left(1 - A_t^p\right) \odot F_t. \tag{2}$$

Only Equation 2 and, for new class sets, one mask-group refinement are incurred per additional output.

3.3 Implementation and Playback on MAXIM

The pipeline is implemented in Python with ONNX Runtime (GPU) for cartoonization and Ultralytics for segmentation, and processes real 360° driving footage recorded with a vehicle-mounted camera (3840×1920 equirectangular frames at 24 fps). Long offline runs are chunked with completion markers and are resumable after interruption. The resulting presets load directly into MAXIM [30, 29], whose Unity-based 360° video-see-through display and vehicle-compensated head tracking [19] present the selected reality through an HMD, with the cabin rendered as virtual vehicle graphics. Because every preset is a per-pixel translation of the same captured ride, the media inherits the footage's physically plausible ego-motion, and in the intended live configuration the visual stream and the vehicle's motion are congruent by construction rather than by authored alignment.

4 Technical Evaluation

We evaluate the pipeline on a one-minute segment of urban driving footage (3840×1920, 24 fps, 1,439 frames) on a desktop with an NVIDIA RTX 3090, with segmentation every second frame (confidence 0.35) and cartoonization at short-side 1280. The evaluation characterizes a proof of concept rather than a deployable real-time loop.

Table 1: Temporal stability of the road-user gating mask over 300 consecutive frames. Consecutive-frame IoU is reported as mean±SD, and dropout events are frame pairs with IoU below 0.3.

Gating mask	IoU $(t,t{+}1)\uparrow$	Dropout events $\downarrow$
Per-frame segmentation	0.572 ± 0.27	18.7%
Flow-warped EMA (ours)	$\mathbf{0.659} \pm 0.26$	**12.0%**
Relative change	+15%	−36%

4.1 Qualitative Results

Figure 3 compares the four presets we focus on in this paper, namely *Original*, *full_toon*, *road_users_toon*, and *road_users_real*. The remaining presets are produced by the same pipeline and differ only in their gating class set (subsection 3.1). The magnified crops show the intended selectivity. The appearance domain switches exactly at the refined mask boundary, without halos of cartoon texture bleeding into photorealistic regions and without photorealistic remnants inside cartoonized objects.

4.2 Temporal Stability of Gating Masks

The gating mask decides which pixels appear cartoonized, so any frame-to-frame change in the mask is directly visible in the output. When a detection briefly drops out, the object loses its cartoon appearance for a frame and then regains it, and the viewer sees the object flicker between the two looks. We quantify stability as the IoU between binarized gating masks of consecutive frames over 300 frames of the road-user group (Table 1). Flow-warped EMA stabilization (Equation 1, production settings) raises the mean consecutive-frame IoU from 0.572 to 0.659, a 15% rela-

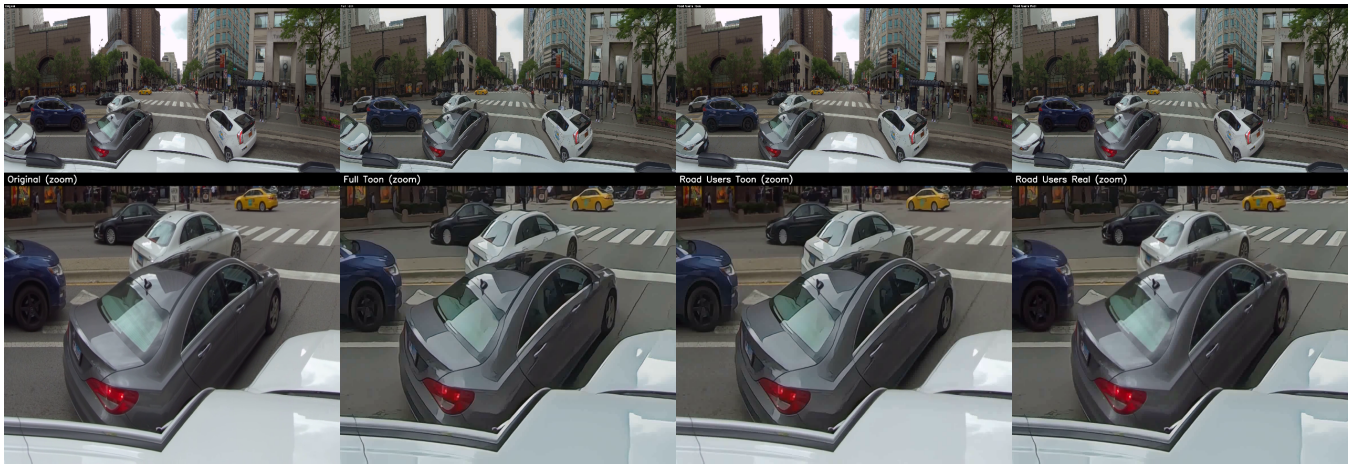


Figure 3: Four representative presets on one frame (top row full view, bottom row magnified crop). From left, *Original*, *full_toon*, *road_users_toon*, and *road_users_real*. Cartoonization is gated precisely along the road-user boundary, with the taxi cartoonized while its surroundings remain photorealistic in *road_users_toon* and the assignment inverted in *road_users_real*.

tive improvement, and reduces mask dropout events from 18.7% to 12.0% of frames, a 36% relative reduction in the more perceptually consequential measure. The absolute values reflect a difficult scene with small, frequently occluded instances entering and leaving the panoramic view, and this mask-level treatment complements temporally consistent translation models, as residual flicker within the cartoon domain remains a property of the frame-wise cartoonizer. For the in-car experience these numbers mean that a chosen reality stays attached to objects for longer stretches of the ride, with fewer moments in which a nearby car or pedestrian abruptly changes appearance and interrupts the immersion.

4.3 Boundary Refinement between Appearance Domains

Figure 4 contrasts three boundary treatments on the same frame and gating mask. This comparison serves as an ablation of a design choice. While the composited results can appear similar in a single still crop, the alpha-map insets show that the edge-aware variant satisfies the requirements of cross-domain composition: a boundary that follows the object silhouette, a fully opaque interior, and a narrow transition band. Violating them has consequences that unfold over time in driving videos. With a hard mask, the boundary inherits the pixel-level noise of the detector, so the seam jitters from frame to frame as the object moves. With Gaussian feathering (kernel about 5% of frame height), the soft halo extends past the object and drags cartoon texture across the moving background, and the blurred interior means the object is never fully cartoonized even in its own *toon* preset. Our edge-aware refinement avoids both failure modes with a single operation on the mask, so its cost also does not grow with the number of presets that share the group. Since these artifacts appear only when the scene moves, boundary quality is best judged on moving footage rather than stills.

4.4 Throughput

Profiled on the evaluation footage, the three shared stages cost approximately 418 ms per frame in total, of which cartoonization accounts for 310 ms, optical flow for 68 ms, and segmentation for 40 ms on the frames where it runs, while each additional preset adds only alpha compositing (1 ms) and, per new class set, one mask-group refinement (18 ms), both executed on the RTX 3090 GPU. Sharing the expensive stages keeps multi-preset authoring possible when needed. The same decomposition also show which stage blocks real-time use. Mask refinement and compositing together stay under 20 ms per output, and implementations of segmentation and optical flow are 15-25 fps, so the gap to real-time rates is concentrated in full-frame cartoonization.

5 Discussion

5.1 Toward Real-Time On-Road Deployment

The intended deployment is a live loop in which the vehicle's camera stream is cartoonized continuously during the ride. The relevant quantities to implement this are single-stream throughput and camera-to-HMD latency, which is critical for visuo-vestibular comfort in moving vehicles [22, 3]. The stage timings in subsection 4.4 leave cartoonization (310 ms) as the only stage far from real time. It can be accelerated by restricting the processed region to the view the passenger currently faces, such as the front view instead of the full 360° frame, or by lowering the processing resolution and attaching a super-resolution stage to restore output quality. Both options trade coverage or fidelity for speed, and such trades act directly on the achievable frame rate, because in a pipelined cascade the slowest stage determines the frame rate while caching of intermediate results lets latency approach the cost of that stage rather than the sum of all stages. Real-time operation thus becomes a matter of tuning these trade-offs, and faster cartoonization models should eventually close the gap without them.

5.2 Native Visual–Motion Congruence and Situation Awareness

A defining property of translating the ride itself is that visual motion and vehicle motion agree by construction. In-car XR content otherwise requires careful alignment of virtual trajectories with the road path [22, 3, 7, 9] or deliberate exploitation of insensitivity to motion gains [28, 13]. Cartoonized reality sidesteps this authoring burden. Decelerations, turns, and bumps are present in the media because they are present in the footage, and unexpected vehicle behavior such as stopping at a light arrives with its visual explanation attached, supporting situation awareness and mitigating the perceived risk of occluded reality in in-car HMD use [11]. A passenger who cannot see why the vehicle brakes hard before a crosswalk must either tolerate the uncertainty or interrupt the experience to check reality. In our system the braking is never unexplained, because the pedestrian on the crossing is part of the media itself, whether cartoonized or photorealistic, so no separate explanatory content and no self-interruption are required. Selectivity then tunes

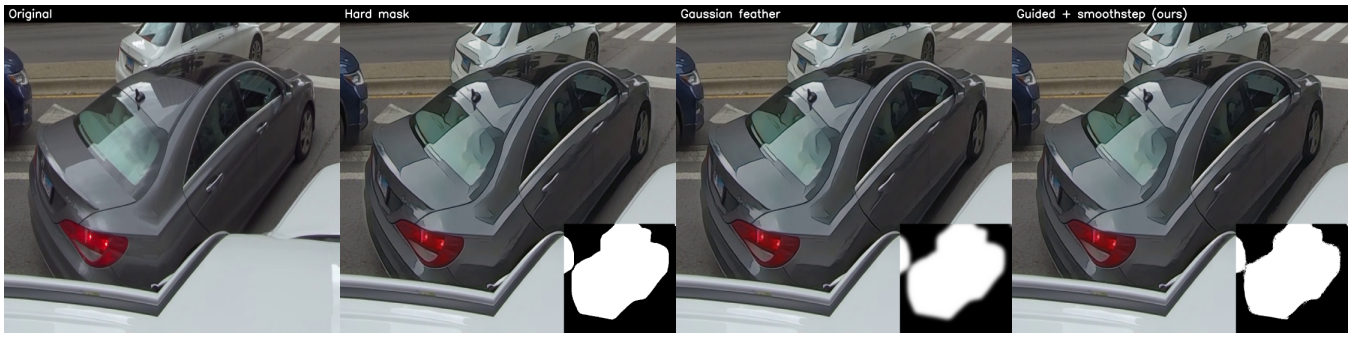


Figure 4: Why the boundary treatment matters (magnified crops, alpha maps as insets). A hard mask ends the cartoon region abruptly along a jagged detection silhouette, leaving a visible seam between the two looks. Gaussian feathering smooths that seam but spreads the cartoon appearance beyond the object and leaves its interior only partially cartoonized. Our guided filtering with smoothstep re-sharpening follows the true silhouette, keeps the interior fully cartoonized, and confines the blend to a narrow band. The alpha maps are the primary evidence, as differences that look subtle in a still frame accumulate under motion.

the awareness–immersion balance [20], with *road_users_real* preserving photorealistic fidelity where interpretive precision matters most, *full_toon* maximizing transformation at the cost of semantic clarity, and intermediate presets populating the continuum. Which preset serves which passenger context is an empirical question that single-pass generation of comparable presets is designed to make studyable.

5.3 From Passive Viewing to Active Scenarios

The current system yields a watchable, mediated ride, a passive experience in which the passenger observes a selectively cartoonized world. Recent in-car XR research demonstrates how rich active experiences can be in fully virtual in-car settings, each pursuing its own experiential goals, as contextual cues hand the passenger agency over locomotion within the physical ride [12], vehicle forces are remapped into diegetic feedback [14], built-in actuators extend the experience multisensorily [31], and tolerable motion gains decouple the virtual camera from the vehicle [28].

One way to bring such activity to the cartoonized ride is to let virtual content enter it. Placing interactive elements over the ride-derived world, for example animated characters inhabiting the cartoonized city or objectives attached to real intersections, corresponds to augmented virtuality on the reality–virtuality continuum [24], with the mediated drive serving as the base world rather than being replaced by fully authored content. Cross-reality blending research further shows that real elements can cross into a virtual world selectively rather than through full passthrough [8], and this is the same selective compositing our pipeline performs. In the vehicle, such policies would let the cabin, the passenger's hands, or an approaching cyclist blend in and out of the cartoonized ride according to interaction and safety demands.

5.4 Limitations

Several limitations qualify our results. All models operate on the equirectangular projection, whose distortion grows toward the poles, so segmentation and cartoonization degrade near the zenith and nadir and the effective cartoon granularity varies with latitude. Polar spatial consistency would require projection-aware processing such as per-face cubemap or viewport-based inference with seam-aware blending. The evaluation is technical, and the perceptual effects of the presets on comfort, situation awareness, sickness, and enjoyment await user studies in the vehicle, for which the MAXIM deployment provides the apparatus. Our footage spans a single urban route in daytime, and robustness across weather, lighting, and sparse scenes remains to be established. The presets also inherit the COCO vocabulary of the segmentation model, which covers road users and key infrastructure but not, for instance, storefronts or vegetation as distinct classes, and panoptic or open-vocabulary segmentation would extend selectivity to scene "stuff". Finally, mask-level stabilization does not eliminate flicker within the cartoonized domain itself, and combining our gating with temporally consistent cartoonization models is a natural next step.

6 Conclusion

We presented *Cartoonize the Ride*, a segmentation-gated pipeline that turns real 360° driving footage into selectively cartoonized reality presets for in-car modified reality, generating ten presets in a single pass while flow-warped mask smoothing and edge-aware boundary refinement keep the seam between cartoon and photograph stable. Because the media is a translation of the ride itself, it natively preserves the visual–motion congruence and road context on which comfortable in-car XR depends, and its class-selectivity provides a concrete instrument for balancing immersion against situation awareness. As a proof of concept, the remaining gap to a live loop is concentrated in the cartoonization stage, and the stage-level analysis suggests that a continuously cartoonized ride, chosen per passenger, per class, and per moment, is an engineering horizon rather than a distant one.

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